

VANSH KUMAR SINGH

+91-9599758037 ◇ vanshkumar841@gmail.com ◇  LinkedIn ◇  Github ◇  X

Bachelor of Technology in Mathematics and Computing, Indian Institute of Technology Delhi (2024)

EDUCATION

YEAR	DEGREE/BOARD	INSTITUTE
2020-2024	B.Tech., Mathematics and Computing	Indian Institute of Technology Delhi

WORK EXPERIENCE

Rootstock | Software Development Engineer

June, 2024 - Present

Customer Refund Workbench

- Built **Customer Refund**, UI+services: AR memos to AP vendor pay, **AR/AP/GL postings** and sub-ledger tie-out
- Architected **5-stage async chain** debit→AP→apply→pay, **savepoints/rollbacks**; AR/AP consistent if a stage fails
- Designed **two-pointer credit matching** + **partial-fill**; blocks **over-apply** on multi-memo refunds, split-credit paths
- Implemented **atomic deactivation** on 3 records; **4-attempt poll** before customer/vendor inactive, no stranded AR/AP

Bank Book (Bank Reconciliation)

- Built **Bank Reconciliation**, UI+services matching daily bank statement lines to AR, AP, and GL book transactions
- Engineered **auto-match engine** with **4-strategy cascade** (check no.→payref→IBAN→invoice) and **sum validation**
- Extended reconciliation for **vendor direct debits** via **IBAN API lookup** and **open AR deposits** by invoice number
- Built **CSV import pipeline (5MB)** with **savepoint-backed rollback** and **chunked writes** under platform limits

INTERNSHIP EXPERIENCE

Accenture | Software Development Engineer Intern

June, 2023 - July, 2023

- Built a blogging platform using **React** and **Cloudflare Workers**, optimizing HTTP request handling for **low latency**
- Implemented **Zod** for JSON schema validation and **type inference**, ensuring robust data integrity across both ends
- Optimized slow **PostgreSQL** queries with **Prisma ORM** and **connection pooling**, improving **query efficiency**
- Implemented secure **signed JWT authentication** system with **cookies**, improving session management and security

Avioliitto Enterprises | Software Development Engineer Intern

May, 2022 - July, 2022

- Developed a peer-to-peer **wallet payment** system using **MERN** stack, securing transfers with JWT and cookies
- Ensured **atomicity** and **data consistency**, preventing double spending and concurrency issues during transactions
- Integrated **transaction notifications** to keep users informed, enhancing trust and the overall user experience
- Designed a user-friendly **React-based UI** for seamless transactions, clear record-keeping and **simple navigation**

PROJECTS

Excalidraw Clone | Real-Time Collaborative Canvas

GitHub

- Built **Canvas 2D** with drag preview and deletes, syncing ops through a dedicated Node WebSocket beside Next.js
- Added join handling with DB snapshot plus server-validated **discriminated ops** into per-room in-memory state
- Handled edit conflicts with shape versions and **baseVersion**, returning op-rejected plus server geometry for resync
- Persisted collaborative canvas JSON in PostgreSQL per room after each op, creating rooms lazily on first **slug** join

Zapier Clone | Workflow Automation Platform

GitHub

- Built **webhook-triggered automation** using Postgres outbox, Kafka, and workers to decouple slow tasks reliably
- Implemented **ordered workflow steps** with templates chaining trigger payload and prior step outputs dynamically
- Validated trigger and action JSON using **Zod**, enabling integrations without schema migrations or runtime errors
- Exposed **Zap APIs** on Next.js using Prisma transactions and Postgres constraints for multi-table consistency checks

Crypto Exchange | Matching Engine

GitHub

- Built **event-driven trade flow** with Next.js APIs publishing to Redis and matcher workers processing orders in flow
- Implemented **in-memory LIMIT/MARKET matching** with best-price checks, partial fills and slippage controls
- Delivered **order book and trade updates** via Redis Pub/Sub to WebSockets with per-symbol subscriptions support
- Added **Prometheus /metrics** and queue push/consume counters for reliable **service monitoring and alerting**

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, JavaScript, TypeScript
- **Backend & Data:** Node.js, Express.js, Next.js, REST APIs, Prisma ORM, PostgreSQL, Redis, Apache Kafka, WebSockets, JWT, OAuth (NextAuth), Zod
- **Frontend:** React, Next.js, Tailwind CSS, responsive UI
- **Cloud & DevOps:** Amazon Web Services (AWS), Docker, Kubernetes, Git, GitHub Actions, CI/CD, Turborepo, Monorepo, Prometheus, Grafana, event-driven architecture, outbox pattern, distributed systems